Kirana Store Profit and Inventory Analytics: Project Report

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Abstract

This project shows how a small kirana store can find the cash locked in its stock and decide what to reorder, stop ordering and clear. It combines real BigBasket product and price data (42,555 clean products) with a simulated 600-product store, analysed in Python and presented in a two-page Power BI dashboard. About 45% of the store's Rs 55.3 lakh inventory sits in overstock and slow-moving items, and 39 of the most profitable (class A) products are overstocked while 8 others need reordering now. The deliverables are a cleaned dataset, a Jupyter notebook, three action lists and the dashboard, all published on GitHub.

Introduction and Problem Statement

Small shopkeepers usually decide how much to buy from experience, so some items pile up while others run out. Excess stock locks up cash and can expire, and a stockout loses sales and customers.

The problem statement of this project is: how can a small business owner see which products earn the most profit, how much cash is stuck in stock, and what to order next? Three questions follow from it:

1. Which products and categories earn the most profit?
2. How much money is tied up in overstock and slow-moving stock?
3. Which products should be reordered, which should stop being ordered, and which should be cleared at a discount?

Objectives

The project aims to turn raw product data into decisions a shopkeeper can act on.

- Combine and clean a large multi-file retail dataset
- Build a realistic kirana store model with cost, demand and stock
- Measure profit by product and category

- Classify stock as healthy, overstock, dead or slow, or reorder now
- Rank products with ABC analysis so attention goes to the products that matter most
- Present the results in an interactive Power BI dashboard with clear action lists

Tools and Technologies

The whole workflow runs on free tools.

Tool	Used for
Python 3.13 (pandas, numpy, matplotlib)	Loading, cleaning, simulating and analysing the data, and drawing charts
VS Code with Jupyter notebook	Writing and running the analysis step by step
Power BI Desktop	Interactive two-page dashboard
Git and GitHub	Version control and publishing the project

Dataset Description

The source is the BigBasket product dataset from Kaggle: 157 Excel files in 10 category folders, 57,013 product rows and 11 columns (category, sub-category, sub-sub-category, product link, EAN code, image link, brand, SKU name, SKU size, MRP and a product description). It gives product names and the maximum retail price (MRP) only, so it has no cost, stock or sales data.

That is why the project mixes real and simulated fields:

Real (from BigBasket)	Simulated (generated for the model)
Product name, category, sub-category, brand, MRP	Margin, cost price, daily demand, units sold, current stock, lead time, reorder level

Data Preparation

Cleaning reduced 57,013 raw rows to 42,555 usable products in six steps.

1. Read all 157 Excel files and join them into one table of 57,013 rows

2. Convert MRP from text such as "₹2,000" to a number
3. Remove 3,921 duplicate products using the product link, leaving 53,092 rows
4. Remove rows with missing or zero MRP (11,315 were missing), leaving 42,555 rows
5. Drop columns not needed for profit and inventory analysis (image link, description, size, EAN code) and fill 3 missing brands with "Unknown"
6. Save the cleaned table so the 157 files need not be read again

Price was not estimated for missing values, because filling it with an average would invent the base of every profit figure.

Kirana Store Model and Simulation

A real kirana stocks a few hundred everyday products, not 42,000, so a 600-product store was built from the cleaned data.

First, only kirana-relevant categories were kept: beverages, foodgrains, oil and masala, bakery and dairy, eight snack sub-categories and five cleaning sub-categories. Party items, toys, car care, bathroom ware, frozen food and mithai were removed, along with products priced above Rs 1,000. This left 16,104 products. A stratified sample then picked 600 with fixed quotas: snacks 260, cleaning 150, beverages 100, foodgrains 60 and bakery and dairy 30. A fixed random seed (42) makes the result repeatable.

The missing business fields were then simulated.

Field	How it was generated
Margin	Random value between 8% and 30% of MRP
Cost price	MRP × (1 − margin)
Daily demand	Skewed random value (lognormal) around a category average, lower for costlier products
Units sold in 90 days	Random count based on daily demand × 90
Current stock	Between 3 and 120 days of demand
Lead time	2 to 7 days
Reorder level	Daily demand × (lead time + 3 safety days)

Derived measures: profit per unit = MRP − cost price; revenue = units sold × MRP; profit = units sold × profit per unit; stock value = stock × cost price; days of cover = stock ÷ daily demand.

Analysis and Findings

Over 90 days the model store earns Rs 19.5 lakh profit on Rs 1.04 crore revenue, but holds Rs 55.3 lakh in stock.

Measure	Value
Revenue (90 days)	Rs 1.04 crore
Profit (90 days)	Rs 19.5 lakh (18.8% margin)
Inventory value at cost	Rs 55.3 lakh
Stock beyond 45 days of demand	Rs 20.9 lakh (38% of inventory)

Snacks and branded foods earn the most profit at Rs 9.2 lakh, about 47% of the total. Beverages follow with Rs 3.5 lakh, then cleaning and household with Rs 3.2 lakh, foodgrains with Rs 2.8 lakh and bakery and dairy with Rs 0.8 lakh. The top single product is RENTIO Heat N Eat Jain Chhole, which sold 2,504 units for Rs 34,555 profit.

Every product was then given a stock status. Overstock means more than 90 days of stock with sales above the median. Dead or slow stock means more than 90 days of stock with sales below the median. Reorder now means stock at or below the reorder level.

Stock status	Products	Stock value	Share of inventory
Healthy	411	Rs 30.2 lakh	54.6%
Overstock (fast-selling)	73	Rs 17.9 lakh	32.3%
Dead or slow stock	76	Rs 7.0 lakh	12.7%
Reorder now	40	Rs 0.3 lakh	0.5%

About 45% of inventory value is in overstock and dead or slow stock. Most of it is overstock of products that do sell, so the first fix is smaller orders, not discounts.

ABC Analysis

ABC analysis shows that the products earning the most profit also hold the most excess stock. It ranks products by profit and groups them: class A gives about 70% of profit, class B the next 20% and class C the last 10%.

Class	Products	Stock value	Share of inventory
A	197	Rs 35.7 lakh	64%
B	169	Rs 11.9 lakh	21%
C	234	Rs 7.8 lakh	14%

Crossing the classes with stock status gives the main insight of the project. Class A holds Rs 13.9 lakh of overstock across 39 products, and 8 other class A products are at or below their reorder level. Class A also carries Rs 13.5 lakh of the Rs 20.9 lakh excess stock, about 65%. At the other end, 44 class C products are dead or slow stock worth Rs 2.2 lakh, which can be cleared with little risk.

In short, ordering is not matched to demand: the best products are both overbought and, in some cases, running out.

Action Plan

The analysis ends in three lists a shopkeeper can act on, saved as an Excel file with one sheet each and shown on page 2 of the dashboard.

List	Rule	Products	What to do
Reorder now	Stock at or below reorder level (A: 8, B: 15, C: 17)	40	Order enough for 30 days of demand, class A first
Stop ordering	Class A with more than 90 days of fast-selling stock	39	Pause orders; about Rs 7.9 lakh sits beyond 45 days of demand
Clearance	Class C dead or slow stock	44	Discount to free about Rs 2.2 lakh of cash

Power BI Dashboard and Version Control

The dashboard has two pages and reads the final 600-product table directly.

- **Overview:** four KPI cards (revenue Rs 1.04 crore, profit Rs 19.5 lakh, inventory value Rs 55.3 lakh, excess stock Rs 20.9 lakh), category and ABC class slicers, a category profit bar chart, a stock status donut, an ABC class stacked column chart and a top 10 products table
- **Action Plan:** the reorder, stop-ordering and clearance tables

Slicers make the page interactive: choosing a category or ABC class updates every visual. The card values (1.95M profit, 10.37M revenue, 5.53M inventory, 2.09M excess stock) and the stock status split were checked against the Python results and match.

The notebook, final dataset, action plan Excel file, Power BI file and a README are published in the GitHub repository kirana-profit-inventory-analytics. The large raw dataset is excluded to keep the repository light.

Limitations

The findings show how the method works, not how a real shop performs, because the core business data is simulated.

- Cost price, demand, stock and lead time are generated, so the Rs 20.9 lakh excess stock and the 45% figure describe the model
- Margins were random, so differences in margin between categories are not meaningful
- Stock was spread evenly between 3 and 120 days of demand, which is more even than a real shop
- Revenue of about Rs 1.15 lakh a day is high for a small store; demand can be scaled down without changing the method
- BigBasket MRP is used in place of a kirana's own selling prices
- Only one 90-day window is covered, so seasonality and festivals are not captured

Conclusion and Future Scope

The project turns a large retail dataset into a working decision tool: it measures profit, finds where cash is stuck, and ends in clear reorder, stop-ordering and clearance lists. Its main contribution is combining ABC analysis with stock status, which shows that the

most profitable products need the most careful ordering.

The same notebook and dashboard can run on a real shop's data with little change. Future work includes:

- Replacing simulated fields with a real shop's billing and stock records
- Forecasting demand for each product to set better reorder levels
- Adding expiry dates and supplier details to the stock rules
- Publishing the dashboard through Power BI Service with scheduled refresh